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MATH 455: Partial Differential Equations

Assignment #5 - Laplace's Equation
Due Thursday, November 5, 2026

1. (14 points) Let $a, b > 0$ and $g \in \mathrm{PC}^1[0, a]$ be given. Find the solution $u(x, y)$ of

$$\Delta u = 0$$

on the rectangle $\Omega = [0, a] \times [0, b]$, with

$$\frac{\partial u}{\partial x}(0, y) = \frac{\partial u}{\partial x}(a, y) = 0$$

for $0 < y < b$ and

$$\frac{\partial u}{\partial y}(x, 0) = 0 \qquad \text{and} \qquad u(x, b) = g(x)$$

for $0 < x < a$.
Hint: It might be easiest to use sinh and cosh in some places rather than real exponentials.

Solution —

$$\Delta u = 0 \implies \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Since this is a linear PDE, we proceed with separation of variables, assuming the solution takes the (separated) form $u(x, y) = X(x)Y(y)$

$$\implies X''(x)Y(y) = -X(x)Y''(y)$$
$$\implies \frac{X''(x)}{X(x)} = -\frac{Y''(y)}{Y(y)} = \lambda$$

This equation takes the form $F(x) = G(y)$ for all $x, y \implies$ both must be equal to some constant λ (fix $x = x_0$, and then $F(x_0) = G(y)$ for all $y \implies G(y)$ is constant $\implies F(x)$ is constant since they're equal to each other). This yields the two ordinary differential equations

$$\begin{cases} X'' - \lambda X = 0 \\ Y'' + \lambda Y = 0 \end{cases}$$

whose solutions depend on the sign of λ.

For $\lambda < 0$:

$$X(x) = c_0 \cos\left(\sqrt{-\lambda}x\right) + c_1 \sin\left(\sqrt{-\lambda}x\right) \qquad Y(y) = c_2 e^{\sqrt{-\lambda}y} + c_3 e^{-\sqrt{-\lambda}y}$$

For $\lambda = 0$:

$$X(x) = Ax + B \qquad Y(y) = Cy + D$$

For $\lambda > 0$:

$$X(x) = c_4 e^{\sqrt{\lambda}x} + c_5 e^{-\sqrt{\lambda}x} \qquad Y(y) = c_6 \cos\left(\sqrt{\lambda}y\right) + c_7 \sin\left(\sqrt{\lambda}y\right)$$

We apply the boundary conditions to $u(x, y) = X(x)Y(y)$:

$$\begin{aligned} \frac{\partial u}{\partial x}(0, y) = 0 &\implies X'(0)Y(y) = 0 \\ &\implies X'(0) = 0 \quad (\text{For } Y(y) \neq 0) \end{aligned}$$

$$\begin{aligned} \frac{\partial u}{\partial x}(a, y) = 0 &\implies X'(a)Y(y) = 0 \\ &\implies X'(a) = 0 \quad (\text{For } Y(y) \neq 0) \end{aligned}$$

We use these results to constrain the unknown constants in our separated solutions:

For $\lambda < 0$:

$$\begin{aligned} X'(0) = 0 &\implies \sqrt{-\lambda}c_1 = 0 \\ &\implies \boxed{c_1 = 0} \\ X'(a) = 0 &\implies -\sqrt{-\lambda}c_0 \sin\left(\sqrt{-\lambda}a\right) = 0 \\ &\implies \sin\left(\sqrt{-\lambda}a\right) = 0 \quad (\text{for nontrivial solution with } c_0 \neq 0) \\ &\implies \sqrt{-\lambda}a = n\pi \quad \text{for } n \in \mathbb{Z} \\ &\implies \boxed{\lambda = -\left(\frac{n\pi}{a}\right)^2} \end{aligned}$$

For $\lambda = 0$:

$$X'(0) = 0, X'(a) = 0 \implies \boxed{A = 0}$$

For $\lambda > 0$:

$$\begin{aligned}
X'(0) = 0 &\implies \sqrt{\lambda}c_4 - \sqrt{\lambda}c_5 = 0 \\
&\implies \boxed{c_4 = c_5} \\
X'(a) = 0 &\implies \sqrt{\lambda}c_4 e^{\sqrt{\lambda}a} - \sqrt{\lambda}c_5 e^{-\sqrt{\lambda}a} = 0 \\
&\implies \sqrt{\lambda}c_4 \left(\underbrace{e^{\sqrt{\lambda}a} - e^{-\sqrt{\lambda}a}}_{\neq 0} \right) = 0 \quad (\text{using } c_4 = c_5) \\
&\implies \boxed{c_4 = 0}
\end{aligned}$$

This leads us to general separated solutions:

For $\lambda < 0$:

$$u(x, y) = \cos\left(\frac{n\pi x}{a}\right)\left(A_n e^{\frac{n\pi y}{a}} + B_n e^{-\frac{n\pi y}{a}}\right)$$

For $\lambda = 0$:

$$u(x, y) = \alpha y + \beta$$

For $\lambda > 0$:

$$u(x, y) = 0$$

Hence the general vertically-constrained solution is the superposition of all these separated solutions:

$$u(x, y) = \alpha y + \beta + \sum_{n=1}^{\infty} \cos\left(\frac{n\pi x}{a}\right)\left(A_n e^{\frac{n\pi y}{a}} + B_n e^{-\frac{n\pi y}{a}}\right)$$

We proceed to apply the horizontal boundary conditions.

$$\frac{\partial u}{\partial y}(x,0) = 0 \implies \alpha + \sum_{n=1}^{\infty} \frac{n\pi}{a} \cos\left(\frac{n\pi x}{a}\right) \left(A_n e^{\frac{n\pi y}{a}} - B_n e^{-\frac{n\pi y}{a}}\right)\Bigg|_{(x,0)} = 0$$

$$\implies \alpha + \sum_{n=1}^{\infty} \frac{n\pi}{a}(A_n - B_n) \cos\left(\frac{n\pi x}{a}\right) = 0$$

Hence α is the constant term and each $n\pi(A_n - B_n)/a$ is the Fourier coefficient of the cosine expansion of $f(x) = 0$ on $x \in [0, a]$. Since the average value of this function is 0 everyone, we have that $\boxed{\alpha = 0}$. Also,

$$\implies \frac{n\pi(A_n - B_n)}{a} = \frac{2}{a} \int_0^a (-\alpha) \cos\left(\frac{n\pi x}{a}\right) dx$$

$$\implies (A_n - B_n) = -\frac{2\alpha}{n\pi} \int_0^a \cos\left(\frac{n\pi x}{a}\right) dx = 0$$

$$\implies \boxed{A_n = B_n}$$

Hence we can write the solution as

$$u(x,y) = \beta + 2\sum_{n=1}^{\infty} C_n \cos\left(\frac{n\pi x}{a}\right) \cosh\left(\frac{n\pi y}{a}\right)$$

We apply the final boundary condition

$$u(x,b) = g(x) \implies \beta + 2\sum_{n=1}^{\infty} C_n \cos\left(\frac{n\pi x}{a}\right) \cosh\left(\frac{n\pi b}{a}\right) = g(x)$$

$$\implies g(x) = \beta + \sum_{n=1}^{\infty} \left[\underbrace{2C_n \cosh\left(\frac{n\pi b}{a}\right)}_{\zeta_n}\right] \cos\left(\frac{n\pi x}{a}\right)$$

$$\implies g(x) = \beta + \sum_{n=1}^{\infty} \zeta_n \cos\left(\frac{n\pi x}{a}\right)$$

Which is the cosine-expansion of $g(x)$ on $[0, a]$, with β being the average value of $g(x)$ on $[0, a]$. Hence

$$\boxed{\beta = \frac{1}{a} \int_0^a g(x) dx}$$

And

$$\zeta_n = \frac{2}{a}\int_0^a g(x)\cos\left(\frac{n\pi x}{a}\right)dx$$
$$\implies 2C_n\cosh\left(\frac{n\pi b}{a}\right) = \frac{2}{a}\int_0^a g(x)\cos\left(\frac{n\pi x}{a}\right)dx$$
$$\implies \boxed{C_n = \frac{1}{a}\operatorname{sech}\left(\frac{n\pi b}{a}\right)\int_0^a g(x)\cos\left(\frac{n\pi x}{a}\right)dx}$$

In the general solution

$$\boxed{u(x,y) = \beta + 2\sum_{n=1}^{\infty} C_n\cos\left(\frac{n\pi x}{a}\right)\cosh\left(\frac{n\pi y}{a}\right)}$$

2. (16 points) Let $a > 0$ be given and find (using polar coordinates) the solution $u(r,\theta)$ of Laplace's equation

$$\Delta u = 0$$

on the wedge

$$\Omega = \left\{(r,\theta) : 0 \le r < a, 0 \le \theta \le \frac{\pi}{3}\right\}$$

with

$$u(r,0) = u\left(r,\frac{\pi}{3}\right) = 0$$

for $0 \le r < a$ and

$$u(a,\theta) = a^2$$

for $0 \le \theta \le \frac{\pi}{3}$. In particular, make sure to only find solutions that remain bounded as $r \to 0$.

Solution — Using polar coordinates conveniently transforms the (non-uniform) boundary of the wedge to a uniform boundary (a rectangle) in polar space. Hence, we must also express the laplace equation in polar form. We apply the change-of-variables

$$x(r,\theta) = r\cos\theta \qquad y(r,\theta) = r\sin\theta$$

And express

$$u(x,y) \implies u(r,\theta)$$

keeping in mind that r and θ are both functions of x and y. We proceed to find $\frac{\partial^2 u}{\partial x^2}$ and $\frac{\partial^2 u}{\partial y^2}$ in terms of θ and r.

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta}\frac{\partial \theta}{\partial x}$$

To find $\frac{\partial r}{\partial x}$ we need r as a function of x and y : $r(x,y)$. We use

$$\begin{aligned} r^2 = x^2 + y^2 &\implies 2r\frac{\partial r}{\partial x} = 2x \\ &\implies \frac{\partial r}{\partial x} = \frac{x}{r} \\ &\implies \boxed{\frac{\partial r}{\partial x} = \cos\theta} \end{aligned}$$

To find $\frac{\partial \theta}{\partial x}$ we need $\theta(x,y)$. We use

$$\begin{aligned} \tan\theta = \frac{y}{x} &\implies \sec^2\theta\frac{\partial \theta}{\partial x} = -\frac{y}{x^2} \\ &\implies \frac{\partial \theta}{\partial x} = -\cos^2\theta\frac{y}{x^2} \\ &\implies \frac{\partial \theta}{\partial x} = -\cos^2\theta\frac{r\sin\theta}{(r\cos\theta)^2} \\ &\implies \boxed{\frac{\partial \theta}{\partial x} = -\frac{\sin\theta}{r}} \end{aligned}$$

Hence

$$\frac{\partial u}{\partial x} = \cos\theta\frac{\partial u}{\partial r} - \frac{\sin\theta}{r}\frac{\partial u}{\partial \theta}$$

And

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x}\left(\cos\theta\frac{\partial u}{\partial r} - \frac{\sin\theta}{r}\frac{\partial u}{\partial \theta}\right) \\ &= \underbrace{\frac{\partial(\cos\theta)}{\partial x}}_{\text{A}}\frac{\partial u}{\partial r} + \cos\theta\underbrace{\frac{\partial}{\partial x}\frac{\partial u}{\partial r}}_{\text{B}} - \underbrace{\frac{\partial(\frac{\sin\theta}{r})}{\partial x}}_{\text{C}}\frac{\partial u}{\partial \theta} - \frac{\sin\theta}{r}\underbrace{\frac{\partial}{\partial x}\frac{\partial u}{\partial \theta}}_{\text{D}} \end{aligned}$$

Simplifying these four pieces:

A:

$$\frac{\partial(\cos\theta)}{\partial x} = \frac{\partial}{\partial x}\left(\frac{x}{\sqrt{x^2+y^2}}\right)$$

$$= \frac{\sqrt{x^2+y^2} - \frac{x^2}{\sqrt{x^2+y^2}}}{x^2+y^2}$$

$$= \frac{r - r\cos^2\theta}{r^2}$$

$$= \frac{\sin^2\theta}{r}$$

B: Note that $\partial u/\partial r$ is some function of r, θ, so let's temporarily define it as the dummy function $\phi(r, \theta)$. Hence,

$$\frac{\partial}{\partial x}\frac{\partial u}{\partial r} = \frac{\partial}{\partial x}\phi(r,\theta)$$

$$= \frac{\partial\phi}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial\phi}{\partial\theta}\frac{\partial\theta}{\partial x}$$

$$= \frac{\partial^2 u}{\partial r^2}\frac{\partial r}{\partial x} + \frac{\partial^2 u}{\partial\theta\partial r}\frac{\partial\theta}{\partial x}$$

$$= \cos\theta\frac{\partial^2 u}{\partial r^2} - \frac{\sin\theta}{r}\frac{\partial^2 u}{\partial\theta\partial r}$$

C:

$$\frac{\partial(\frac{\sin\theta}{r})}{\partial x} = \frac{\partial}{\partial x}\left(\frac{y}{x^2+y^2}\right)$$

$$= \frac{-2xy}{(x^2+y^2)^2}$$

$$= \frac{-2r^2\sin\theta\cos\theta}{r^4}$$

$$= \frac{-2\sin\theta\cos\theta}{r^2}$$

$$= -\frac{\sin 2\theta}{r^2}$$

D: Note that $\partial u/\partial\theta$ is some function r, θ, so let's also temporarily define it as the dummy function $\phi(r,\theta)$. Then

$$\frac{\partial}{\partial x}\frac{\partial u}{\partial \theta} = \frac{\partial}{\partial x}\phi(r,\theta)$$

$$= \frac{\partial\phi}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial\phi}{\partial\theta}\frac{\partial\theta}{\partial x}$$

$$= \frac{\partial^2 u}{\partial r\partial\theta}\frac{\partial r}{\partial x} + \frac{\partial^2 u}{\partial\theta^2}\frac{\partial\theta}{\partial x}$$

$$= \cos\theta\frac{\partial^2 u}{\partial r\partial\theta} - \frac{\sin\theta}{r}\frac{\partial^2 u}{\partial\theta^2}$$

Hence

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} =& \frac{\sin^2\theta}{r}\frac{\partial u}{\partial r} + \cos\theta\left[\cos\theta\frac{\partial^2 u}{\partial r^2} - \frac{\sin\theta}{r}\frac{\partial^2 u}{\partial\theta\partial r}\right] \\ &+ \frac{\sin 2\theta}{r^2}\frac{\partial u}{\partial\theta} - \frac{\sin\theta}{r}\left[\cos\theta\frac{\partial^2 u}{\partial r\partial\theta} - \frac{\sin\theta}{r}\frac{\partial^2 u}{\partial\theta^2}\right]\end{aligned}$$

$$\frac{\partial^2 u}{\partial x^2} = \cos^2\theta\frac{\partial^2 u}{\partial r^2} + \frac{\sin^2\theta}{r}\frac{\partial u}{\partial r} - \frac{\sin 2\theta}{r}\frac{\partial^2 u}{\partial r\partial\theta} + \frac{\sin^2\theta}{r^2}\frac{\partial^2 u}{\partial\theta^2} + \frac{\sin 2\theta}{r^2}\frac{\partial u}{\partial\theta}$$

An analogous process yields

$$\frac{\partial^2 u}{\partial y^2} = \sin^2\theta \frac{\partial^2 u}{\partial r^2} + \frac{\cos^2\theta}{r}\frac{\partial u}{\partial r} + \frac{\sin 2\theta}{r}\frac{\partial^2 u}{\partial r \partial \theta} + \frac{\cos^2\theta}{r^2}\frac{\partial^2 u}{\partial \theta^2} - \frac{\sin 2\theta}{r^2}\frac{\partial u}{\partial \theta}$$

Hence

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial u^2}{\partial \theta^2}$$

So the polar Laplacian equation must be

$$\boxed{\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0}$$

Now we proceed to find the general solution of this differential equation, using the same separation of variables method as before. We assume $u(r,\theta)$ takes the form $R(r)\Theta(\theta)$. Then

$$R''(r)\Theta(\theta) + \frac{R'(r)\Theta(\theta)}{r} + \frac{R(r)\Theta''(\theta)}{r^2} = 0$$

Dividing through by $R(r)\Theta(\theta)$ yeilds

$$\frac{R''(r)}{R(r)} + \frac{R'(r)}{rR(r)} + \frac{\Theta''(\theta)}{r^2\Theta(\theta)} = 0$$

$$\implies \underbrace{r^2\frac{R''(r)}{R(r)} + r\frac{R'(r)}{R(r)}}_{F(r)} = \underbrace{-\frac{\Theta''(\theta)}{\Theta(\theta)}}_{G(\theta)} = \mu$$

The left and right are functions of r only, and θ only, respectively. So our differential equal that we assume works for all r, θ implies this equation, which, also, supposedly works for all r, θ. And since $F(r) = G(\theta)$ for all r, θ, then both sides must be equal to the same constant (fix $\theta = \theta_0$ and we get $F(r) = G(\theta_0)$ for all r, the definition of a constant function). We call this constant (the separation constant) μ.

$$\implies \begin{cases} r^2R'' + rR' - \mu R = 0 \\ \Theta'' + \mu\Theta = 0 \end{cases}$$

Since the radial equation is non-trivial, let us solve the easier angular equation in its entirety first. By the characteristic equation, we obtain

For $\mu > 0$:

$$\Theta(\theta) = c_0\cos\left(\sqrt{\mu}\theta\right) + c_1\sin\left(\sqrt{\mu}\theta\right)$$

For $\mu = 0$:

$$\Theta(\theta) = c_2 + c_3\theta$$

For $\mu < 0$:

$$\Theta(\theta) = c_4 e^{\sqrt{-\mu}\theta} + c_5 e^{-\sqrt{-\mu}\theta}$$

We apply the angular Neumann boundary condition

$$\begin{aligned} & u(r,0) = u\left(r, \frac{\pi}{3}\right) = 0 \\ \implies & R(r)\Theta(0) = 0 \implies \Theta(0) = 0 \quad \text{(For nontrivial } R(r)\text{)} \\ \implies & R(r)\Theta\left(\frac{\pi}{3}\right) = 0 \implies \Theta\left(\frac{\pi}{3}\right) = 0 \quad \text{(For nontrivial } R(r)\text{)} \end{aligned}$$

For $\mu > 0$:

$$\begin{aligned} \Theta(0) = 0 &\implies \boxed{c_0 = 0} \\ \Theta\left(\frac{\pi}{3}\right) = 0 &\implies c_1 \sin\left(\sqrt{\mu}\frac{\pi}{3}\right) = 0 \\ &\implies \sin\left(\sqrt{\mu}\frac{\pi}{3}\right) = 0 \quad (\text{for } c_1 \neq 0) \\ &\implies \frac{\pi\sqrt{\mu}}{3} = n\pi \quad \text{For } n \in \mathbb{Z} \\ &\implies \boxed{\mu = (3n)^2} \end{aligned}$$

For $\mu = 0$:

$$\begin{aligned} \Theta(0) = 0 &\implies \boxed{c_2 = 0} \\ \Theta\left(\frac{\pi}{3}\right) = 0 &\implies \boxed{c_3 = 0} \end{aligned}$$

For $\mu < 0$:

$$\begin{aligned} \Theta(0) = 0 &\implies \boxed{c_4 = -c_5} \\ \Theta\left(\frac{\pi}{3}\right) = 0 &\implies c_4 e^{\sqrt{-\mu}\frac{\pi}{3}} + c_5 e^{-\sqrt{-\mu}\frac{\pi}{3}} = 0 \\ &\implies c_4\left(e^{\sqrt{-\mu}\frac{\pi}{3}} - e^{-\sqrt{-\mu}\frac{\pi}{3}}\right) = 0 \quad (\text{substituting } c_5 = -c_4) \\ &\implies 2c_4 \underbrace{\sinh\left(\sqrt{-\mu}\frac{\pi}{3}\right)}_{\neq 0} = 0 \\ &\implies \boxed{c_4 = 0} \end{aligned}$$

We obtain trivial solutions for everything except the $\mu > 0$ case, so that case is the only contributor of angular eigenfunction solutions:

$$\Theta_n(\theta) = c_1 \sin(3n\theta)$$

Let us now solve the radial equation, given that we know $\mu = (3n)^2$. We must hence solve

$$r^2 R'' + rR' - (3n)^2 R = 0$$

This is linear but has **non**-constant coefficients, making it impossible to solve using the characteristic equation. Instead, this has the form of a Cauchy-Euler differential equation, which suggests the solution guess $R(r) = r^k$. Plugging in yields

$$\begin{aligned} & k(k-1)r^k + kr^k - (3n)^2 r^k = 0 \\ \implies & k(k-1) + k - (3n)^2 = 0 \\ \implies & k^2 - (3n)^2 = 0 \\ \implies & k = \pm 3n \end{aligned}$$

Hence the general solution is

$$R(r) = c_6 r^{3n} + c_7 r^{-3n}$$

We discard the $c_7 r^{-3n}$ because we only want solutions that remain bounded as $r \to 0$.

$$R(r) = c_6 r^{3n}$$

Hence the general bounded solution satisfying the angular boundary conditions is

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n r^{3n} \sin(3n\theta)$$

We now apply the radial boundary condition to resolve the A_n's.

$$\begin{aligned} & u(a, \theta) = a^2 \\ \implies & \sum_{n=1}^{\infty} A_n a^{3n} \sin(3n\theta) = a^2 \end{aligned}$$

To find A_n, we pattern-match with a known sin-Fourier expansion, for instance something like

$$\sum_{n=1}^{\infty} \zeta_n \sin\left(\frac{n\pi x}{L}\right) \qquad \text{(sin-expansion of a general } f(x) \text{ on } [0, L])$$

We manipulate our equation into this form

$$\implies \sum_{n=1}^{\infty} A_n a^{3n} \sin\left(\frac{n\pi\theta}{\pi/3}\right) = a^2$$

$\implies A_n a^{3n}$ are the coefficients of the sin-expansion of $f(\theta) = a^2$ on $[0, \pi/3]$.

$$\begin{aligned}
\implies A_n a^{3n} &= \frac{2}{\pi/3} \int_0^{\frac{\pi}{3}} a^2 \sin(3n\theta)\, d\theta \\
\implies A_n &= \frac{6a^{2-3n}}{\pi} \int_0^{\frac{\pi}{3}} \sin(3n\theta)\, d\theta \\
&= -\frac{6a^{2-3n}}{\pi} \frac{1}{3n} \Big[\cos(3n\theta) \Big]_{\theta=0}^{\theta=\frac{\pi}{3}} \\
&= -\frac{6a^{2-3n}}{\pi} \frac{1}{3n} \left[\cos(n\pi) - 1 \right] \\
&= -\frac{6a^{2-3n}((-1)^n - 1)}{3n\pi} \\
&= \begin{cases} 0 & \text{if } n \text{ even} \\ \frac{4a^{2-3n}}{n\pi} & \text{if } n \text{ odd} \end{cases}
\end{aligned}$$

$$\implies u(r,\theta) = \frac{4a^2}{\pi} \sum_{n=1}^{\infty} \frac{r^{3n} \sin(3n\theta)}{n a^{3n}} \quad \text{for odd } n$$

$$\implies u(r,\theta) = 0 \quad \text{for even } n$$

Hence, to obtain all the non-zero terms exactly, we hop over the odd numbers $1, 3, 5 \cdots$ by hopping over $2n+1$ starting at $n = 0$. Hence

$$u(r,\theta) = \frac{4a^2}{\pi} \sum_{n=0}^{\infty} \frac{r^{3(2n+1)} \sin(3(2n+1)\theta)}{(2n+1) a^{3(2n+1)}}$$

$$\boxed{u(r,\theta) = \frac{4a^2}{\pi} \sum_{n=0}^{\infty} \frac{1}{2n+1} \left(\frac{r}{a}\right)^{6n+3} \sin((6n+3)\theta)}$$

3. (20 points) Let $0 < R_1 < R_2$ be given and g be a PC^1 function. Find the single-valued solution of Laplace's equation on the annulus

$$\Omega = \{(r,\theta) : R_1 < r < R_2, -\pi \le \theta < \pi\}$$

with the boundary conditions

$$u(R_1,\theta) = 0 \qquad \text{and} \qquad u(R_2,\theta) = g(\theta)$$

for $-\pi \le \theta < \pi$. In particular, use the periodic boundary conditions

$$u(r,-\pi) = u(r,\pi) \qquad \text{and} \qquad \frac{\partial u}{\partial \theta}(r,-\pi) = \frac{\partial u}{\partial \theta}(r,\pi)$$

for $R_1 < r < R_2$ in order to compute the solution.

Solution — As derived before, the general solution $u(r,\theta) = R(r)\Theta(\theta)$ to the polar Laplacian equation is constructed of the following general angular eigenfunctions:

$$\begin{aligned}
&\textbf{For } \mu > 0: \quad \Theta(\theta) = c_0 \cos\left(\sqrt{\mu}\theta\right) + c_1 \sin\left(\sqrt{\mu}\theta\right) \\
&\textbf{For } \mu = 0: \quad \Theta(\theta) = c_2 + c_3\theta \\
&\textbf{For } \mu < 0: \quad \Theta(\theta) = c_4 e^{\sqrt{-\mu}\theta} + c_5 e^{-\sqrt{-\mu}\theta}
\end{aligned}$$

We proceed to find the boundary-condition-applied general angular solution and use the resulting μ-constraint to solve the radial equation.

$$\begin{aligned}
u(r,-\pi) = u(r,\pi) &\implies R(r)\Theta(-\pi) = R(r)\Theta(\pi) \\
&\implies \Theta(-\pi) = \Theta(\pi) \quad \text{(For nontrivial } R(r) \neq 0\text{)}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial u}{\partial \theta}(r,-\pi) = \frac{\partial u}{\partial \theta}(r,\pi) &\implies R(r)\Theta'(-\pi) = R(r)\Theta'(\pi) \\
&\implies \Theta'(-\pi) = \Theta'(\pi) \quad \text{(For nontrivial } R(r) \neq 0\text{)}
\end{aligned}$$

For $\mu > 0$:

$$\begin{aligned}
\Theta(-\pi) = \Theta(\pi) &\implies c_0 \cos\left(-\pi\sqrt{\mu}\right) + c_1 \sin\left(-\pi\sqrt{\mu}\right) = c_0 \cos\left(\pi\sqrt{\mu}\right) + c_1 \sin\left(\pi\sqrt{\mu}\right) \\
&\implies -c_1 \sin(\pi\sqrt{\mu}) = c_1 \sin(\pi\sqrt{\mu}) \\
&\implies 2c_1 \sin(\pi\sqrt{\mu}) = 0 \\
&\implies c_1 = 0 \quad \text{OR} \quad \sin(\pi\sqrt{\mu}) = 0 \\
&\qquad\qquad \implies \mu = n^2 \quad (\text{for } n \in \mathbb{Z})
\end{aligned}$$

$$\begin{aligned}
\Theta'(-\pi) = \Theta'(\pi) &\implies -c_0\sqrt{\mu}\sin(-\pi\sqrt{\mu}) = -c_0\sqrt{\mu}\sin(\pi\sqrt{\mu}) \\
&\implies 2c_0\sqrt{\mu}\sin(\pi\sqrt{\mu}) = 0 \\
&\implies c_0 = 0 \quad \text{OR} \quad \sin(\pi\sqrt{\mu}) = 0 \\
&\qquad\qquad \implies \mu = n^2 \quad (\text{for } n \in \mathbb{Z})
\end{aligned}$$

For any nontrivial solution, at least one of c_0, c_1 would be non-zero, forcing one of the $\mu = n^2$ conditions above. Hence, any nontrivial solution must have $\boxed{\mu_n = n^2}$.

For $\mu = 0$:

$$\Theta(-\pi) = \Theta(\pi) \implies c_2 - \pi c_3 = c_2 + \pi c_3$$
$$\implies \boxed{c_3 = 0}$$

$$\Theta'(-\pi) = \Theta'(\pi) \implies c_3 = c_3 \quad \text{(no useful constraint)}$$

For $\mu < 0$:

$$\begin{aligned}
\Theta(-\pi) = \Theta(\pi) &\implies c_4 e^{-\pi\sqrt{-\mu}} + c_5 e^{\pi\sqrt{-\mu}} = c_4 e^{\pi\sqrt{-\mu}} + c_5 e^{-\pi\sqrt{-\mu}} \\
&\implies c_4 + c_5 e^{2\pi\sqrt{-\mu}} = c_4 e^{2\pi\sqrt{-\mu}} + c_5 \quad \text{(multiplying through by } e^{\pi\sqrt{-\mu}}) \\
&\implies e^{2\pi\sqrt{-\mu}}(c_5 - c_4) = (c_5 - c_4) \\
&\implies (c_5 - c_4)[e^{2\pi\sqrt{-\mu}} - 1] = 0 \\
&\implies \boxed{c_4 = c_5} \quad \text{OR} \quad e^{2\pi\sqrt{-\mu}} = 1 \\
&\qquad\qquad\qquad\qquad \implies 2\pi\sqrt{-\mu} = 0 \\
&\qquad\qquad\qquad\qquad \text{(No solutions for the case } \mu < 0)
\end{aligned}$$

$$\begin{aligned}
\Theta'(-\pi) = \Theta'(\pi) &\implies c_4\sqrt{-\mu}e^{-\pi\sqrt{-\mu}} - c_4\sqrt{-\mu}e^{\pi\sqrt{-\mu}} = c_4\sqrt{-\mu}e^{\pi\sqrt{-\mu}} - c_4\sqrt{-\mu}e^{-\pi\sqrt{-\mu}} \\
&\implies c_4 e^{-\pi\sqrt{-\mu}} - c_4 e^{\pi\sqrt{-\mu}} = c_4 e^{\pi\sqrt{-\mu}} - c_4 e^{-\pi\sqrt{-\mu}} \\
&\implies 2c_4 e^{-\pi\sqrt{-\mu}} = 2c_4 e^{\pi\sqrt{-\mu}} \\
&\implies 2c_4\left(e^{\pi\sqrt{-\mu}} - e^{-\pi\sqrt{-\mu}}\right) = 0 \\
&\implies 4c_4 \underbrace{\sinh\left(e^{\pi\sqrt{-\mu}}\right)}_{\neq 0} = 0 \\
&\implies \boxed{c_4 = 0}
\end{aligned}$$

Hence, the only nontrivial angular solutions we find are for $\mu = 0$, yielding a constant solution

$$\Theta_0(\theta) = c_4$$

and for $\mu > 0$ yeilding:

$$\Theta_n(\theta) = c_0 \cos(n\theta) + c_1 \sin(n\theta)$$

Now we can solve the radial equation with the knowledge that $\mu = n^2$. Note that, unlike in the previous problem, we *do* have nontrivial (constant) solutions coming from the $\mu =$

$0 \implies n = 0$ case above. Hence, we must consider the $n = 0$ case separately is because it changes the structure of the radial differential equation:

$$r^2 R'' + rR' - n^2 R = 0$$

For $n > 0$: This is a Cauchy-Euler equation, so we guess $R(r) = r^k$ as before.

$$\begin{aligned}
&\implies k(k-1)r^k + kr^k - n^2 r^k = 0 \\
&\implies k(k-1) + k - n^2 = 0 \\
&\implies k^2 = n^2 \\
&\implies k = \pm n
\end{aligned}$$

So the general radial solution is

$$R_n(r) = c_6 r^n + c_7 r^{-n}$$

every component of which remains bounded on our domain (an annulus).

For $n = 0$: The equation becomes

$$r^2 R'' + rR' = 0$$

This looks similar to the original Cauchy-Euler equation, so we guess $R(r) = r^k$ again:

$$\begin{aligned}
&\implies k(k-1)r^k + kr^k = 0 \\
&\implies k(k-1) + k = 0 \\
&\implies k^2 = 0 \\
&\implies k = 0 \quad \text{(repeated root)}
\end{aligned}$$

Which yields only $R(r) = c_8$ (a constant solution). Because this is a second-order equation, we need two linearly independent solutions, so we use another method to solve the equation.

$$\begin{aligned}
&r^2 R'' + rR' = 0 \\
&\implies rR'' + R' = 0 \\
&\implies \frac{d}{dr}(rR') = 0 \\
&\implies rR' = C \\
&\implies R' = C/r \\
&\implies R(r) = C\ln(r) + B
\end{aligned}$$

Hence the general solution for the $n = 0$ case can be expressed as

$$R_0(r) = c_8 + c_9 \ln r$$

In total, we have now found all the general solutions satisfying the angular boundary constraints:

$$\begin{aligned}
\textbf{For } \mu > 0 \ (n > 0): &\quad u(r,\theta) = R_n\Theta_n = (c_6 r^n + c_7 r^{-n})(c_0\cos(n\theta) + c_1\sin(n\theta)) \\
\textbf{For } \mu = 0 \ (n = 0): &\quad u(r,\theta) = R_0\Theta_0 = (c_8 + c_9\ln r)c_4 \\
\textbf{For } \mu < 0: &\quad u(r,\theta) = 0
\end{aligned}$$

Cleaning the constants up:

$$\begin{aligned}
\textbf{For } \mu > 0 \ (n > 0): &\quad u(r,\theta) = A_n r^n\cos(n\theta) + B_n r^n\sin(n\theta) + C_n r^{-n}\cos(n\theta) + D_n r^{-n}\sin(n\theta) \\
\textbf{For } \mu = 0 \ (n = 0): &\quad u(r,\theta) = \alpha + \beta\ln r \\
\textbf{For } \mu < 0: &\quad u(r,\theta) = 0
\end{aligned}$$

Hence, the full general solution satisfying the angular boundary conditions is

$$u(r,\theta) = \alpha + \beta\ln r + \sum_{n=1}^{\infty}\left[\left(A_n r^n + C_n r^{-n}\right)\cos(n\theta) + \left(B_n r^n + D_n r^{-n}\right)\sin(n\theta)\right]$$

We now apply the radial boundary conditions:

$$\begin{aligned}
&u(R_1,\theta) = 0 \\
\implies &\alpha + \beta\ln R_1 + \sum_{n=1}^{\infty}\left[\left(A_n R_1^n + C_n R_1^{-n}\right)\cos(n\theta) + \left(B_n R_1^n + D_n R_1^{-n}\right)\sin(n\theta)\right] = 0
\end{aligned}$$

which is of the Fourier series form

$$\alpha + \beta\ln R_1 + \sum_{n=1}^{\infty}\left[\eta_n\cos(n\theta) + \zeta_n\sin(n\theta)\right] = 0$$

This is the Fourier-expansion of the function $f(\theta) = 0$ on $\theta \in [-\pi,\pi]$, with constant term $\alpha + \beta\ln R_1$. Notice that this constant term must be zero because the function that is being Fourier expanded is zero itself (so its average value is also zero). This implies $\boxed{\alpha = -\beta\ln R_1}$. Because this is the Fourier expansion of the zero-function, then the Fourier coefficients must all be zero. In particular,

$$\begin{aligned}
&\eta_n = 0 \\
&\zeta_n = 0
\end{aligned}$$

$$\begin{aligned}
\implies A_n R_1^n + C_n R_1^{-n} &= 0 \\
\implies C_n R_1^{-n} &= -A_n R_1^n \\
\implies C_n &= -A_n R_1^{2n}
\end{aligned}$$

$$\begin{aligned}
\implies B_n R_1^n + D_n R_1^{-n} &= 0 \\
\implies D_n R_1^{-n} &= -B_n R_1^n \\
\implies D_n &= -B_n R_1^{2n}
\end{aligned}$$

Hence, the general solution with all angular boundary conditions and the inner radial boundary condition is

$$u(r,\theta) = -\beta \ln R_1 + \beta \ln r + \sum_{n=1}^{\infty} \left[A_n \left(r^n - \left(\frac{R_1^2}{r} \right)^n \right) \cos(n\theta) + B_n \left(r^n - \left(\frac{R_1^2}{r} \right)^n \right) \sin(n\theta) \right]$$

$$\boxed{u(r,\theta) = \beta \ln \left(\frac{r}{R_1} \right) + \sum_{n=1}^{\infty} \left(r^n - \left(\frac{R_1^2}{r} \right)^n \right) \left[A_n \cos(n\theta) + B_n \sin(n\theta) \right]}$$

We now apply the outer radial boundary condition on R_2.

$$\begin{aligned} & u(R_2,\theta) = g(\theta) \\ \implies & \beta \ln \left(\frac{R_2}{R_1} \right) + \sum_{n=1}^{\infty} \left(R_2^n - \left(\frac{R_1^2}{R_2} \right)^n \right) \left[A_n \cos(n\theta) + B_n \sin(n\theta) \right] = g(\theta) \end{aligned}$$

This is the Fourier-expansion of $g(\theta)$ on $\theta \in [-\pi, \pi]$, where

$$\begin{aligned} \beta \ln \left(\frac{R_2}{R_1} \right) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} g(\theta) d\theta \\ \implies & \boxed{\beta = \frac{1}{2\pi \ln(R_2/R_1)} \int_{-\pi}^{\pi} g(\theta) d\theta} \end{aligned}$$

And

$$\begin{aligned} A_n \left(R_2^n - \left(\frac{R_1^2}{R_2} \right)^n \right) &= \frac{1}{\pi} \int_{-\pi}^{\pi} g(\theta) \cos(n\theta) d\theta \\ \implies & \boxed{A_n = \frac{R_2^n}{\pi (R_2^{2n} - R_1^{2n})} \int_{-\pi}^{\pi} g(\theta) \cos(n\theta) d\theta} \end{aligned}$$

And

$$\begin{aligned} B_n \left(R_2^n - \left(\frac{R_1^2}{R_2} \right)^n \right) &= \frac{1}{\pi} \int_{-\pi}^{\pi} g(\theta) \sin(n\theta) d\theta \\ \implies & \boxed{B_n = \frac{R_2^n}{\pi (R_2^{2n} - R_1^{2n})} \int_{-\pi}^{\pi} g(\theta) \sin(n\theta) d\theta} \end{aligned}$$